

Yash Mahesh Kamerkar

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SUMMARY

Software Engineer and AI Engineer with 3 years of experience building production-grade **microservices** and large language model (**LLM**) applications across logistics and enterprise technology organizations. Delivered backend services handling 45K daily transactions in **Java** and **Spring Boot** at Dell Technologies, and built an **LLM**-powered customer support assistant at Universal Courier using **Python**, **LangChain**, and **OpenAI API** that automated 420 delivery inquiry resolutions per month and cut first-response time from 6 hours to 12 minutes. Proficient in retrieval-augmented generation (**RAG**), **vector databases**, **REST API** development, and cloud-native deployment on Amazon Web Services (**AWS**) and Microsoft **Azure**.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), TypeScript, SQL, Bash

AI/ML & Generative AI: LangChain, LangGraph, OpenAI API, Anthropic Claude API, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Prompt Engineering, Hugging Face Transformers, BERT, scikit-learn, PyTorch, vector databases

Frameworks & Libraries: FastAPI, Flask, Spring Boot, Spring MVC, Hibernate ORM, Pandas, NumPy, React.js, JUnit 5, Mockito

Databases: PostgreSQL, MySQL, Oracle, MongoDB, Pinecone, ChromaDB, Redis

Cloud & DevOps: Amazon Web Services (AWS), Microsoft Azure, Azure OpenAI Service, Docker, Kubernetes (K8s), Jenkins, GitHub Actions, Terraform, CI/CD Pipelines

Tools & Methodologies: Git, GitHub, JIRA, Postman, Maven, Gradle, Linux, REST APIs, Microservices Architecture, Agile/Scrum

EXPERIENCE

Uber

Aug 2026 – Present

Software Engineer | AI Engineer

USA

- Architected a **gRPC**-based trip-event aggregation **microservice** in **Java** and **Spring Boot**, consolidating driver telemetry from 4 upstream location feeds and cutting duplicate event processing incidents from 9 to 2 per week.
- Engineered a retrieval-augmented generation (**RAG**) pipeline using **LangChain**, **OpenAI API**, and **Pinecone** to power an internal driver policy assistant, cutting policy lookup time from 15 minutes to under 50 seconds across 12,000 compliance documents.

Universal Courier

Jul 2023 – Aug 2024

Software Engineer | AI Engineer

India

- Integrated 3 third-party carrier **REST APIs** (UPS, FedEx, USPS) using **Python** and **FastAPI** to consolidate real-time shipment status for 8,000 daily parcels into a unified customer-facing tracking dashboard.
- Designed a delivery **ETA prediction model** using **scikit-learn** trained on 2 years of historical shipment records, reducing average estimation error from 3.4 hours to 1.1 hours across 5 regional delivery zones.
- Built an **LLM**-powered customer support assistant using **LangChain** and **OpenAI API** to handle inbound delivery inquiry tickets, automating resolution of 420 repetitive queries per month and cutting average first-response time from 6hr to 12 min.
- Constructed **REST API** endpoints in **FastAPI** to expose real-time route and package status data to 3 internal logistics dashboards, supporting 150 concurrent dispatcher users during peak morning scheduling windows.
- Developed an **ETL** pipeline in **Python** using **Pandas** and **SQLAlchemy** that ingested and normalized 180 GB of daily shipment records from 6 warehouse management systems, enabling downstream analytics and executive reporting.
- Automated nightly route-optimization batch jobs using **Jenkins** and **Docker** on Microsoft **Azure**, compressing manual scheduling overhead from 4 hours to 35 minutes per dispatch cycle across 3 regional distribution hubs.

Dell Technologies

Sep 2021 – Jul 2023

Software Engineer

India

- Implemented order-management **microservices** in **Java 11** and **Spring Boot 2.x** for Dell's supply chain platform, processing 45,000 daily purchase requests across 6 interconnected service modules.
- Migrated 3 legacy **Oracle PL/SQL** batch jobs to **Python**-based **ETL** pipelines using **Pandas** and **SQLAlchemy**, reducing nightly batch processing time from 6 hours to 2.5 hours.
- Authored **JUnit 5** and **Mockito** test suites for 12 core service modules, eliminating 14 critical regression defects before each production release and adding 1,500 net tested lines across the codebase.
- Deployed containerized **Spring Boot** services via **Docker** on **Kubernetes** clusters, enabling zero-downtime rolling updates across 4 staging and production environments.
- Streamlined **RESTful API** contracts for 3 product catalog endpoints using **Spring MVC** and **Hibernate ORM**, cutting internal portal integration onboarding time from 5 days to 2 days.
- Optimized **PostgreSQL** query performance for 5 high-frequency reporting endpoints by introducing composite indexes and rewriting 3 N+1 query patterns, reducing average database response time from 2.1 seconds to 340 ms.
- Refactored a monolithic inventory reporting module into 2 independently deployable **Spring Boot** services, cutting module build time from 14 minutes to 5 minutes and removing 3 cross-team deployment dependencies.

EDUCATION

Stevens Institute of Technology

Master of Science in Computer Science

Hoboken, NJ

Sep 2024 – May 2026